

# $\sigma$ -Additive Probability on Orthomodular Lattices: A Survey and an Open Problem

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## Abstract

The passage from a finitely coherent assignment of probabilities to a countably additive measure is, in the Boolean case, governed by a single condition (countable additivity) and executed by classical machinery (Carathéodory; Loomis–Sikorski; Stone duality). On a *non-distributive* orthomodular lattice (OML) — the algebra of propositions of a quantum system — this machinery breaks, and the question of when finite coherence forces countable behaviour is open. We survey the problem at the level needed to begin work on it. After fixing the apparatus (orthomodular lattices, the gap between orthogonality and meet-zero, the state/meet-additive/charge ladder, the McDonald–Bimbó dual space), we separate the question into two axes. The *extension* axis is classical (Horn–Tarski feasibility) and fully settled: on  $L(H)$  no state of any kind extends, by an elementary finite construction. The *descent* axis is where the genuinely open question lives, and we locate it precisely. The combinatorial route — seeking the object among concrete,  $\sigma$ -orthocomplete, set-representable OMLs — is **settled and does not produce it**: the interleaving property that would make  $\sigma$ -additivity exceed finite additivity on a concrete non-Boolean lattice is satisfied trivially by the plain product  $\prod_n \text{MO}_2$ , which is concrete,  $\sigma$ -complete, non-Boolean, infinite — and exactly the *segregated* object (non-distributivity confined to finite blocks) that a relational probability theory must exclude. The deeper reason is that a faithful  $\sigma$ -tribe representation has a distributive image, forcing the OML Boolean, so the point-based route is closed for every non-Boolean OML (and the witness  $\prod_n \text{MO}_2$  has, concretely, no two-valued homomorphisms at all); the forcing boundary to Boolean is *subadditivity* of the separating states (Pták–Pulmannová), not  $\sigma$ -additivity. The open problem is therefore necessarily **point-free**: does a non-Boolean,  $\sigma$ -complete OML carry a  $\sigma$ -additive probability that is genuinely relational — the non-Boolean analogue of a localic (point-free) measure — rather than reducible to a point space? The two richly non-distributive poles frame it:  $L(H)$ +Gleason has a  $\sigma$ -additive lattice measure but is the *most* point-ful object, not point-free (the (A)/(B) analysis of §5); the concrete combinatorial objects are segregated. A positive answer is the non-Boolean analogue of localic measure theory; a negative answer, an impossibility theorem closing the cluster. We review what is known and what is ruled out, and close with the directions in which the problem might be approached.

## 1 Introduction

All measurement is finite, yet the frameworks that organise empirical knowledge — probability theory, quantum mechanics — rest on infinite structures. The passage from finite observation to infinite structure is not automatic; it requires a commitment no finite body of evidence can compel. In probability theory the locus of that commitment is *countable additivity*:

$$A_1 \supseteq A_2 \supseteq \cdots, \quad \bigcap_n A_n = \emptyset \quad \implies \quad \mu(A_n) \rightarrow 0.$$

De Finetti [8] held that only finite additivity is empirically grounded; Kolmogorov [24] adopted  $\sigma$ -additivity as an axiom.

The same tension recurs once observations need not be jointly performable. An observation *context* — a set of measurements performable together — is Boolean; gluing contexts along shared observations (a categorical colimit) returns:

contexts mutually compatible  $\Rightarrow$  Boolean algebra,      incompatible  $\Rightarrow$  non-distributive OML

(Gunji et al. [17]; the dichotomy organises the companion paper [25]). The OML is *forced* by the context structure, not postulated. Incompatibility — quantum complementarity, but equally contextual experimental design or interfering sensors — drives the algebra out of the Boolean world. One caveat fixes the scope: for non-quantum examples the incompatibility must be *operationally imposed* (contexts genuinely non-co-realizable, no joint probability space); it cannot arise intrinsically from one classical system’s dynamics, since any finite family of classical observables on a common space jointly distributes (Kolmogorov), hence stays Boolean — contextuality there is failure of a global section (Abramsky–Brandenburger [1]; Fine [12]). The closed subspaces of a Hilbert space  $H$  form the prototype  $L(H)$ , not the premise.

On a non-distributive OML, Carathéodory’s outer measure — needing a Boolean algebra of sets — does not apply, and the rescue is local to  $L(H)$ :

- For  $L(H)$ , Gleason’s rigidity theorem [15] rescues it; Hilbert space is special and no general-OML analogue is known.
- Non-distributivity *per se* does not remove the 2-valued homomorphisms the Boolean engine needs —  $\text{MO}_3$  is non-distributive yet carries them (§2.1, §4) — but it removes the *guarantee*; for  $L(H)$ ,  $\dim \geq 3$  they fail outright (Kochen–Specker [23]).

*Whether enough survive is a question of concreteness, not non-distributivity* — where an OML is *concrete* (Definition 2.7) when its elements can be faithfully modelled as honest *sets of outcomes*, equivalently when it carries an order-determining set of two-valued states (Gudder [16]); “enough survive” is then almost the definition. This is the qualifier on which the open problem turns (§4, §5).

**The organising split.** This note asks the shared question — *when does finite coherence force countable behaviour?* — on an OML. The vague question splits into two axes that the Boolean case fuses, and the survey is built around the split:

| Axis      | Asks                                            | Status                                 |
|-----------|-------------------------------------------------|----------------------------------------|
| Extension | does a state extend to a charge on $S_0(A)$ ?   | settled (on $L(H)$ , <i>none</i> does) |
| Descent   | does the charge concentrate on physical points? | <b>open</b>                            |

**The relational standpoint.** Two established moves motivate the approach:

- De Finetti [8] — probability without a presupposed sample space: coherent commitments about events, not measures on a given  $\Omega$ .
- Birkhoff–von Neumann — propositions as a lattice (the OML of §2), not as subsets of a phase space.

Combining them, we start from the propositions and their entailment order and ask what that relational structure alone determines — the measure-theoretic counterpart of pointless (localic) topology. The dual is then not a space of points but the McDonald–Bimbó space of *filters* (§2.3), and a positive answer to the open problem would be a *point-free*  $\sigma$ -additive probability theory on a non-distributive lattice — the non-Boolean analogue of localic measure theory (§5; developed further in [26]).

## 2 Preliminaries

We collect the lattice-theoretic apparatus; all notions are standard.

## 2.1 Orthomodular lattices, and the orthogonal/meet-zero gap

**Definition 2.1** (Orthocomplemented lattice). A *bounded lattice* is a partially ordered set in which every pair  $a, b$  has a join  $a \vee b$  and meet  $a \wedge b$ , with greatest element  $\mathbf{1}$  and least element  $\mathbf{0}$ . An *orthocomplementation* is a map  $a \mapsto a^\perp$  satisfying, for all  $a, b$ :

- (i)  $a \wedge a^\perp = \mathbf{0}$  and  $a \vee a^\perp = \mathbf{1}$  (complement),
- (ii)  $a^{\perp\perp} = a$  (involution),
- (iii)  $a \leq b \Rightarrow b^\perp \leq a^\perp$  (order-reversing).

**Definition 2.2** (Incomparable; meet-zero; orthogonal). For elements  $a, b$  of an orthocomplemented lattice, three relations of increasing strength:

- *incomparable*:  $a \not\leq b$  and  $b \not\leq a$  — neither entails the other (an order condition);
- *meet-zero*:  $a \wedge b = \mathbf{0}$  — no common refinement;
- *orthogonal*, written  $a \perp b$ :  $a \leq b^\perp$  (equivalently  $b \leq a^\perp$ ) — one entails the other's negation, *decisively incompatible*.

Each implies the one above it ( $a \perp b \Rightarrow a \wedge b = \mathbf{0} \Rightarrow$  incomparable, for  $a, b \notin \{\mathbf{0}, \mathbf{1}\}$ ). For two *distinct atoms*, incomparable and meet-zero coincide — an atom has only  $\mathbf{0}$  below it, so  $a \wedge b \in \{\mathbf{0}, a\}$ , and  $a \wedge b = a$  would force  $a \leq b$ ; orthogonality remains strictly stronger.

*Remark 2.3* (Meet-zero versus orthogonal). The implication  $a \perp b \Rightarrow a \wedge b = \mathbf{0}$  reverses in every Boolean algebra, but fails once the lattice is non-distributive: there meet-zero is strictly weaker than orthogonality. The distinctions developed below all turn on this separation.

**Definition 2.4** (Orthomodular lattice). An orthocomplemented lattice  $A$  is *orthomodular* if

$$a \leq b \implies b = a \vee (a^\perp \wedge b). \quad (1)$$

A *Boolean algebra* is the special case where  $\wedge$  distributes over  $\vee$ ; every Boolean algebra is orthomodular, not conversely. The motivating non-Boolean example is  $L(H)$ , the closed subspaces of a Hilbert space with  $a^\perp$  the orthogonal complement.

**Definition 2.5** (Completeness notions). Two countable-completeness conditions on an OML  $A$  are distinguished throughout, the second strictly weaker. Here  $\bigvee_n a_n$  denotes the *join* (least upper bound) of  $\{a_n\}$  in the order of  $A$ , unique when it exists; “ $\exists \bigvee_n a_n \in A$ ” asserts that it does.

- $A$  is  $\sigma$ -complete if

$$\forall \{a_n\}_{n \in \mathbb{N}} \subseteq A : \quad \exists \bigvee_n a_n \in A$$

(equivalently every countable family has a meet, by  $\bigwedge_n a_n = (\bigvee_n a_n^\perp)^\perp$ );

- $A$  is  $\sigma$ -orthocomplete if

$$\forall \{a_n\}_{n \in \mathbb{N}} \subseteq A \ (a_i \perp a_j \text{ for } i \neq j) : \quad \exists \bigvee_n a_n \in A.$$

The orthogonal families are a subclass of all countable families, so  $\sigma$ -completeness quantifies over a strictly larger domain and is the stronger hypothesis:

$$\{a_n\} \text{ pairwise orthogonal} \subseteq \{a_n\} \text{ arbitrary} \implies \sigma\text{-complete} \Rightarrow \sigma\text{-orthocomplete}.$$

Only the weaker condition is needed below. For a state  $s$ ,  $\sigma$ -additivity is the requirement

$$s\left(\bigvee_n a_n\right) = \sum_n s(a_n) \quad \text{for every countable } \{a_n\} \text{ with } a_i \perp a_j \ (i \neq j),$$

whose left side invokes  $\bigvee_n a_n$  *only* for orthogonal families — so  $\sigma$ -orthocompleteness is exactly the hypothesis that makes it well-posed. Full  $\sigma$ -completeness is the stronger condition manipulated by the Loomis–Sikorski representation (§2.4).

**Example 2.6** ( $\text{MO}_n$ , and the gap made concrete).  $\text{MO}_n$  is the lattice with  $\mathbf{0}$ ,  $\mathbf{1}$ , and  $n$  complementary pairs of incomparable atoms, all sharing only  $\mathbf{0}$  and  $\mathbf{1}$ . The geometric model is  $n$  lines through the origin: each line  $a$  is orthogonal to its perpendicular  $a^\perp$ , while two atoms from distinct pairs are non-perpendicular lines. It is orthomodular and non-distributive ( $n \geq 2$ ): two atoms drawn from *distinct* complementary pairs have

$$a \wedge b = \mathbf{0} \quad (\text{meet only at the origin}) \quad \text{yet} \quad a \not\perp b,$$

so meet-zero  $\neq$  orthogonal already here.  $\text{MO}_3$  is the smallest non-distributive OML.

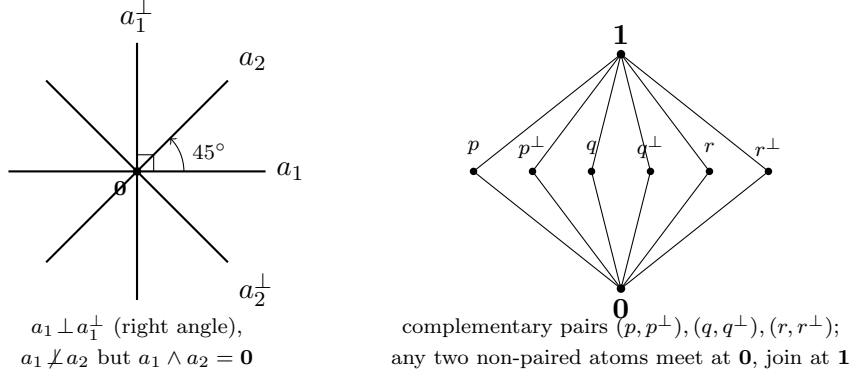


Figure 1:  $\text{MO}_n$  illustrated. *Left*: the geometric model for  $\text{MO}_2$  — four atoms as lines through the origin in  $\mathbb{R}^2$ . The pair  $a_1, a_1^\perp$  meets at a right angle (orthogonal), whereas  $a_1, a_2$  meet at  $45^\circ$ : their lattice meet is still  $\mathbf{0}$  (only the origin in common), yet they are *not* orthogonal — the meet-zero  $\neq$  orthogonal gap. *Right*: the Hasse diagram of  $\text{MO}_3$ , the smallest non-distributive OML: three complementary pairs of incomparable atoms between  $\mathbf{0}$  and  $\mathbf{1}$ .

**Definition 2.7** (OMP; order-determining states; concrete logic). An *orthomodular poset* (OMP) weakens Definition 2.4: joins  $a \vee b$  are required only for *orthogonal* pairs. A set  $S$  of states on  $A$  is *order-determining* if it separates the order, i.e.

$$(s(a) \leq s(b) \text{ for all } s \in S) \implies a \leq b.$$

An OMP is a *concrete logic* (equivalently *set-representable*) if it embeds into  $(\mathcal{P}(X), \subseteq, X \setminus (-))$  for some set  $X$ , preserving existing orthogonal joins — i.e. its propositions are modelled as actual *sets of outcomes*. Concreteness is equivalent to admitting an order-determining set of *two-valued* states (Gudder [16]), and it is the property that separates the open frontier from the settled cases (§5).

*Remark 2.8* (Concreteness does not close the gap). A set-representable OMP needs only closure under complement and *disjoint* union [4]; intersection-closure would force a field of sets, hence Boolean. The lattice meet  $a \wedge b$  is the greatest  $L$ -member inside  $A \cap B$ , so

$$\underbrace{A \cap B = \emptyset}_{\text{orthogonal}} \implies \underbrace{a \wedge b = \mathbf{0}}_{\text{meet-zero}}, \quad \text{but not conversely.}$$

Concreteness does not force the converse:  $\text{MO}_3$  is itself concrete (Definition 2.7; Gudder [16]), so the paradigmatic meet-zero  $\neq$  orthogonal example is already a concrete logic. What closes the gap is a richness/atomicity condition, not concreteness:

$$a \wedge b = \mathbf{0} \implies a \perp b \iff \text{every nonempty } A \cap B \text{ contains a nonzero } L\text{-member,}$$

the abstract form being Tkadlec's *Boolean orthoposet* condition [35] (which does *not* mean Boolean *algebra*).

## 2.2 The state/meet-additive/charge ladder

**Definition 2.9** (State; meet-additive state; charge-extendible). Let  $A$  be an OML. A *state* is a map  $s : A \rightarrow [0, 1]$  with  $s(\mathbf{1}) = 1$  that is additive on *orthogonal* pairs:  $s(a \vee b) = s(a) + s(b)$  whenever  $a \perp b$ ; that is, a probability assignment whose additivity is guaranteed only on decisively incompatible propositions. Three strengthenings, differing in which pairs additivity is demanded on and strictly increasing in general (Example 2.10 on  $\text{MO}_3$ ):

- (1) *State*:  $s(a \vee b) = s(a) + s(b)$  whenever  $a \perp b$ .
- (2) *Meet-additive*:  $s(a \vee b) = s(a) + s(b)$  whenever  $a \wedge b = \mathbf{0}$  (binary, strictly stronger — it reaches the gap pairs of Definition 2.2). We avoid the word “valuation” here: in lattice theory it denotes the modular function  $v(a) + v(b) = v(a \wedge b) + v(a \vee b)$ , a different condition.
- (3) *Charge-extendible*:  $\sum_i s(a_i) \leq 1$  for every pairwise-meet-zero family  $\{a_i\}$  ( $a_i \wedge a_j = \mathbf{0}$ ,  $i \neq j$ ; strictly stronger again). This is the condition necessary for extension to a charge on the dual space (Definition 2.13) and the *extension* condition of §3.

**Example 2.10** (The ladder is strict). On  $\text{MO}_3$  with complementary pairs  $(p, p^\perp), (q, q^\perp), (r, r^\perp)$ , the maximally mixed  $s \equiv \frac{1}{2}$  is meet-additive (clears (2)) yet fails (3) at  $\{p, q, r\}$ :  $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$ . So no state on  $\text{MO}_3$  is charge-extendible.

**Definition 2.11** (Normal and singular states on  $L(H)$ ). On  $A = L(H)$  a state  $s$  is

- *normal* if it is completely additive on every orthogonal family of projections,

$$s\left(\bigvee_{i \in I} p_i\right) = \sum_{i \in I} s(p_i) \quad \text{for all (not only countable) orthogonal } \{p_i\},$$

equivalently  $s(\cdot) = \text{tr}(\rho \cdot)$  for a density operator  $\rho$  (Gleason [15],  $\dim \geq 3$ );

- *singular* (purely) if

$$s(e) = 0 \quad \text{for every finite-rank projection } e.$$

Every state decomposes into a normal and a singular part (Takesaki); the two classes are exactly those distinguished by the arguments of §3.

## 2.3 The McDonald–Bimbó dual space

**Definition 2.12** (Dual space; representation map). The McDonald–Bimbó duality [27] assigns to an OML  $A$  a compact space

$$S_0(A) = (F(A), \subseteq, \perp_A, P(A), T),$$

where  $F(A)$  is the set of *filters* of  $A$ , ordered by  $\subseteq$  and carrying an orthogonality relation  $\perp_A$  and a Stone topology  $T$ , and  $P(A) = \{\uparrow p : p \in A\} \subseteq F(A)$ , with  $\uparrow p = \{a \in A : a \geq p\}$ , is the set of *principal* filters — the “physical points,” the realisation datum (canonical, unlike the non-canonical pure points of the Boolean case). For  $a \in A$  set  $h(a) = \{x \in F(A) : a \in x\}$ ; then  $h$  is an OML isomorphism onto the  $\perp$ -stable clopens  $\text{CO}(S_0(A))^\dagger$ . Define  $\mu(h(a)) = s(a)$ . Here  $h(a)$  is the set of points at which  $a$  holds and  $\mu$  is the state transported onto those sets; the open problem is whether  $\mu$  is preserved under countable operations on them.

**Definition 2.13** (Charge). A *charge* on  $S_0(A)$  is a map  $\nu : \text{CO}(S_0(A)) \rightarrow [0, 1]$  on the *full* Boolean algebra of clopen subsets (not merely the  $\perp$ -stable ones in the image of  $h$ ), with

$$\nu(S_0(A)) = 1, \quad \nu(U \cup V) = \nu(U) + \nu(V) \quad \text{whenever } U \cap V = \emptyset.$$

A state  $s$  *extends to a charge* if some such  $\nu$  satisfies  $\nu(h(a)) = s(a)$  for all  $a \in A$ . Because  $h$  preserves meets, a pairwise-meet-zero family in  $A$  maps to disjoint clopens, so a charge necessarily satisfies  $\sum_i s(a_i) \leq 1$  over such families — the charge-extendibility condition of Definition 2.9(3).

*Remark 2.14* (The duality is finitary). The representation map  $h$  satisfies

$$h\left(\bigvee_{i \leq n} a_i\right) = \left(\bigcup_{i \leq n} h(a_i)\right)^{\perp\perp} \quad (\text{finite}), \quad h\left(\bigvee_n a_n\right) \neq \left(\bigcup_n h(a_n)\right)^{\perp\perp} \quad (\text{countable}) :$$

finitary OML homomorphisms need not preserve infinite suprema. This is the central obstruction the open problem of §5 confronts, and it is self-dual: since  $a \mapsto a^\perp$  is an order anti-isomorphism,  $\bigwedge_n a_n = (\bigvee_n a_n^\perp)^\perp$ , so the failure for countable meets is the De Morgan dual of the failure for countable joins. The rival duality of Cannon–Döring [5] is also finitary and carries no measure; we use McDonald–Bimbó because it carries  $P(A)$  as explicit structure.

## 2.4 The Boolean engine: Stone duality

In the Boolean case the descent equivalence

$$\sigma\text{-additive} \iff \text{the measure concentrates on the physical points}$$

is proved through *Stone duality*, in four steps [26]:

- (S1) a Boolean event algebra  $B$  embeds in the clopen algebra  $\text{CO}(\text{St } B)$  of its Stone space;
- (S2) a charge on  $B$  extends to a Baire measure on  $\text{St } B$  (Carathéodory);
- (S3)  $\sigma$ -additivity  $\iff$  the measure vanishes on meagre Baire sets (Rao–Rao [33]);
- (S4) meagre-null  $\iff$  the measure lives on the ultrafilters closed under countable intersection = the realised points.

Steps (S3)–(S4) are the measure-theoretic face of the *Loomis–Sikorski* representation,

$$\sigma\text{-complete Boolean algebra} \cong (\sigma\text{-tribe of sets}) / (\sigma\text{-ideal}),$$

the countable-join bookkeeping that drives the equivalence. The contrast with the OML case is exact:

| Boolean                                      | OML                       |
|----------------------------------------------|---------------------------|
| Stone duality + Loomis–Sikorski (the engine) | <b>no analogue</b> (§4.3) |

Supplying such an engine, or proving none exists, is the open problem of §5.

*Remark 2.15* (Why the bridge breaks: meet-closed but not join-prime). A “realised” point  $x$  carries two conditions, which Rao–Rao’s “vanishes on meagre sets” makes coincide in the Boolean case:

- (M) closed under countable meets;      (J)  $x \ni \bigvee_n a_n \Rightarrow x \ni a_n$  some  $n$  (off the join-defect).

Non-distributivity separates them. For a principal filter  $x = \uparrow p$ :

- (M) always holds.
- (J) can fail:  $p \leq \bigvee_n a_n$  does not force  $p \leq a_n$ .

In  $L(H)$  this is witnessed by  $a_n = \text{span}(e_n)$  and  $p = \text{span}(e_1 + e_2)$ :

$$p \leq \bigvee_n a_n = H, \quad \text{yet} \quad p \not\leq a_n \text{ for every } n.$$

So  $\uparrow p$  is a realised point lying inside the join-defect  $D = h(\bigvee_n a_n) \setminus \bigcup_n h(a_n)$ . Topologically  $\uparrow p$  is *isolated* in  $S_0(L(H))$ , hence:

$$D \text{ non-meagre (OML)} \quad \text{vs.} \quad D \text{ nowhere dense (Boolean).}$$

This is the finitary obstruction of Remark 2.14 on the dual: the category route (clopens modulo the meagre ideal) fails on the same  $D$  as the measure route, and reduces to the MacNeille completion of §4.3(ii).

### 3 Two axes: extension and descent

The central question — *when does a state  $s$  on an OML  $A$  extend to a  $\sigma$ -additive measure on  $S_0(A)$ ?* — splits into two **independent** axes that the Boolean case fuses. A state  $s$  on  $A$  faces:

|                    | Extension axis                                                                                          | Descent axis                                                                |
|--------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Question</b>    | Does $\mu$ extend to a finitely additive charge on the <i>full</i> Boolean clopen algebra of $S_0(A)$ ? | Once extended, does the measure concentrate on the physical points $P(A)$ ? |
| <b>Governed by</b> | the meet-zero/orthogonal gap (§2.1)                                                                     | $\sigma$ -additivity — needs the missing engine (§2.4)                      |
| <b>Status</b>      | settled:      Horn–Tarski / Pitowsky feasibility; finite case = decidable LP; can fail for every state. | open.                                                                       |

The extension axis is the abstract form of a familiar question: extending a finitely-additive assignment on incompatible propositions to a single joint distribution is exactly the feasibility problem behind the Bell inequalities (Fine [12]: a Bell model exists iff the correlations admit a joint distribution; Pitowsky [30]: the Bell inequalities are the facets of a correlation polytope), and Bell and Kochen–Specker non-classicality are jointly an extension problem for partial Boolean algebras (Budroni–Morchio [2], with the necessary-and-sufficient conditions of Horn–Tarski [21]). What is distinctive here is only that the extension is sought against the dual  $S_0(A)$  rather than within a fixed scenario; the obstruction is the same finitary one, and it is settled. Two observations pin it down.

**Proposition 3.1** (Finite case). *For finite  $A$  the extension condition is a decidable linear program (Horn–Tarski [21] / Pitowsky correlation-polytope feasibility [30]; Budroni–Morchio [2] for the partial-Boolean-algebra formulation); since  $h$  preserves meets, meet-zero elements map to disjoint clopens and a charge must satisfy  $\sum_i s(a_i) \leq 1$  over pairwise-meet-zero families. It can fail for every state (MO<sub>3</sub>, Example 2.10).*

**Proposition 3.2** (Infinite  $L(H)$ , all states). *Let  $\dim H = \infty$ . No state on  $L(H)$  — normal or singular — extends to a charge on  $S_0(L(H))$ .*

*Proof.* The argument embeds a copy of MO<sub>2</sub> in  $L(H)$  whose four atoms are infinite-dimensional, so that the obstruction reaches the singular states as well. Write  $H = H_0 \otimes \mathbb{C}^2$  with  $H_0$  infinite-dimensional, and fix an orthonormal basis  $e, f$  of  $\mathbb{C}^2$ . Set

$$a_1 = H_0 \otimes \mathbb{C}e, \quad a_1^\perp = H_0 \otimes \mathbb{C}f, \quad a_2 = H_0 \otimes \mathbb{C}(e+f), \quad a_2^\perp = H_0 \otimes \mathbb{C}(e-f).$$

These four infinite-dimensional subspaces form a copy of MO<sub>2</sub> in  $L(H)$ :

$$a_1 \perp a_1^\perp, \quad a_2 \perp a_2^\perp, \quad a_i \vee a_i^\perp = H; \quad a_i \wedge a_j = \mathbf{0} \text{ but } a_i \not\perp a_j \text{ (cross pairs).}$$

Orthoadditivity with  $s(\mathbf{1}) = 1$  gives  $s(a_i) + s(a_i^\perp) = 1$  for  $i = 1, 2$ , using only additivity on orthogonal pairs, so it holds for *every* state. The family  $\{a_1, a_1^\perp, a_2, a_2^\perp\}$  is pairwise meet-zero, and since  $h$  preserves meets ( $h(a \wedge b) = h(a) \cap h(b)$ ), its members map to disjoint clopens; hence any charge satisfies  $\sum_i s(a_i) \leq 1$ . The two counts collide:

$$\sum_i s(a_i) = 2 \text{ (orthoadditivity)} \quad \text{versus} \quad \sum_i s(a_i) \leq 1 \text{ (any charge).}$$

The construction is finite ( $n = 4$ );  $\sigma$ -completeness of the lattice plays no role. □

The  $\text{MO}_2$  count is standard (orthoadditivity together with  $s(\mathbf{1}) = 1$ ; see [22]) and  $\sum s(a_i) = 2$  is already state-independent in  $L(\mathbb{C}^2)$ . The role of the *infinite-dimensional* realisation is to extend the obstruction to the singular states ( $s(e) = 0$  on every finite-rank  $e$ ):

| Atoms of $\text{MO}_2$                      | $s(a_i) + s(a_i^\perp)$                   | Reaches                           |
|---------------------------------------------|-------------------------------------------|-----------------------------------|
| lines in $L(\mathbb{C}^2)$ (finite rank)    | $= 1$ only if $s$ charges finite rank     | normal states only                |
| $H_0 \otimes (\cdot)$ , $\dim H_0 = \infty$ | $= 1$ regardless of finite-rank behaviour | <b>all states, incl. singular</b> |

Keeping each atom infinite-dimensional drops the finite-rank precondition of the line argument (Remark 3.3), and that is what closes the extension axis on  $L(H)$  completely.

*Remark 3.3* (An alternative argument for normal states only). A second, technique-disjoint argument reaches the normal states without the  $\text{MO}_2$  family. If  $s(e) > 0$  on some finite-dimensional  $e$ , it is positive on a 2-plane  $e_0$ ; taking  $k$  pairwise-non-orthogonal lines  $p_1, \dots, p_k \subset e_0$  yields  $2k$  pairwise-meet-zero lines with  $s(p_i) + s(p_i^\perp) = s(e_0)$ , so a charge forces

$$k s(e_0) \leq 1 \text{ for all } k \implies s(e_0) = 0,$$

contradicting  $s(e_0) > 0$ . Lifting  $s$  to a functional via Mackey–Gleason / Bunce–Wright [3] ( $B(H)$  has no type  $\text{I}_2$  summand) and applying the Takesaki normal/singular decomposition identifies which states the argument reaches:

|                |                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------|
| <b>reached</b> | nonzero normal part ( $s(e) > 0$ on some finite $e$ ); includes every normal (Gleason) state                |
| <b>missed</b>  | purely singular ( $s(e) = 0$ for all finite $e$ ); e.g. ultrafilter vector states, Calkin-algebra pullbacks |

Proposition 3.2 subsumes the missed case; this argument is recorded because its technique is independent and locates the singular states precisely.

## 4 What is known

The extension axis is settled — on  $L(H)$  no state extends (Proposition 3.2); the descent axis turns on whether the Boolean engine of §2.4 has an OML analogue. Three things bear on that:

1. the Boolean benchmark such an analogue must match (§4.1);
2. the partial OML results, positive and obstructive (§4.2);
3. the routes already known to be blocked (§4.3).

### 4.1 The Boolean benchmark

The conditions an OML engine would have to reproduce. The Kelley–Vladimirov–Pták characterisation (Fremlin [13], Thm 391D) is complete: a Boolean algebra carries a strictly positive  $\sigma$ -additive measure iff it satisfies all three conditions below — each of which loses its meaning off the distributive world.

| Boolean (KVP) condition                 | OML analogue                                                    |
|-----------------------------------------|-----------------------------------------------------------------|
| Dedekind $\sigma$ -completeness         | exists ( $\sigma$ -orthocompleteness)                           |
| weak $(\sigma, \infty)$ -distributivity | <b>none</b> — relies on $\bigwedge$ distributing over $\bigvee$ |
| chargeability                           | <b>none</b> — no structural characterisation on OMLs            |



## 4.2 Positive and obstructive OML results

**Positive** (require structure rich enough to approximate Hilbert-space geometry):

- Gleason [15] —  $L(H)$ ,  $\dim \geq 3$ ;  $\sigma$ -additivity *on the lattice*.
- Bunce–Wright [3] — JBW projection lattices; Chetcuti–Dvurečenskij [6].

**Obstructive** (Boolean machinery cannot be transplanted):

- Pták–Pulmannová [32] — a *finitary* condition (*subadditive*-unitality  $s(a \vee b) \leq s(a) + s(b)$ , no countable joins) already collapses the *lattice* to Boolean (their Thm 1); the *OMposet* version fails (Müller’s set-representable counterexample [28]).
- Navara [29] — regularity does not force  $\sigma$ -additivity. Pták [31] — exotic state spaces via Greechie pasting.

## 4.3 Three routes to a $\sigma$ -OML engine, all unavailable

The descent argument needs an OML analogue of the Boolean engine — a  $\sigma$ -Stone duality, or equivalently a Loomis–Sikorski representation. The three known routes are all unavailable, but not on the same footing: routes (i) and (ii) are closed by *theorem* (the hypothesis that would yield the representation provably forces Boolean, resp. leaves the category), whereas route (iii) is open in the strong sense — no construction is known, and no impossibility theorem rules one out (Remark 4.1). It is route (iii), restated, that is the Open Problem of §5.

- (i) **RDP / effect-algebra.** Loomis–Sikorski holds for  $\sigma$ -MV algebras [10] and monotone  $\sigma$ -effect algebras *with* the Riesz Decomposition Property [11]; but a lattice effect algebra has RDP iff it is MV, and  $\text{OML} \cap \text{MV} = \text{Boolean}$  (Kalmbach [22]), so the hypothesis that would yield the representation collapses the class:

$$\text{OML} + \text{RDP} \iff \text{Boolean}.$$

- (ii) **MacNeille completion.** The MacNeille completion of an OML need not be orthomodular (Harding [18]); one cannot complete to absorb countable joins. The topological version of this route — clopens of  $S_0(A)$  modulo the meagre ideal — is this same completion in another form (the regular-open algebra), and collapses non-distributivity (Remark 2.15).
- (iii)  **$\sigma$ -Stone duality.** None is *known*, and none is *ruled out*: the existing dualities are finitary (McDonald–Bimbó, Remark 2.14) or equational with no set/tribe representation (Freytes [14], on  $\sigma$ -complete OMLs), but no theorem forbids a countable-join-preserving one.

## 4.4 Prior art on point-free quantum probability

Every existing point-free or directed-context construction fails at least one of the two requirements —  $\sigma$ -additivity and a natively non-distributive structure — as tabulated below.

### Gleason tradition

Varadarajan [36], Kalmbach [22], Pták–Pulmannová [32]:  $\sigma$ -additive on a non-distributive OML, but real-valued on a fixed lattice — not point-free.

### Döring 2009

[9] point-free on the spectral presheaf over a directed poset — but provably only *finitely additive*, on a distributive Heyting algebra.

### Bohrification (HLS)

[19, 20] point-free and the internal valuation *is*  $\sigma$ -additive (Scott-continuous) — but factors through an internally *distributive* locale,  $\sigma$ -additive only per commutative context. Non-distributivity is tamed, not carried natively.

### Localic valuations

Vickers [37], Simpson [34], Coquand–Spitters [7]: point-free  $\sigma$ -additive, but on *frames* — distributive by definition.

### OML Stone dualities

McDonald–Bimbó [27] and Cannon–Döring [5]: purely structural, no measure, no  $\sigma$ -version.

Each entry satisfies some of the requirements but not all. The open case is their conjunction, satisfied by no existing construction:

point-free  $\times$   $\sigma$ -additive  $\times$  natively non-distributive  $\times$  directed-system-built.

(The directed-context-yields-non-distributivity construction is not new — it is the shape of Bohrification and of colimit-over-Boolean-contexts generation. Novelty would lie only in the  $\sigma$ -additive  $\times$  point-free  $\times$  native combination.)

*Remark 4.1* (The boundary of the negative results). Two qualifications delimit what is ruled out.

- (i) **No impossibility theorem covers the concrete class.** The known no-go results are finite or two-valued, and the one positive occupant is non-concrete:

$$L(H) \text{ (dim} \geq 3\text{)} : \quad \text{no two-valued states at all (Kochen–Specker [23]),}$$

a fortiori none order-determining. Concreteness is the qualifier on which the open frontier turns.

- (ii) **No  $\sigma$ -Loomis–Sikorski exists**, and this is visible in the dualities: both take infinite joins as a closure, not a union,

$$\text{Cannon–Döring: } \text{cls}(\bigcup S_i); \quad \text{McDonald–Bimbó: } (\bigcup h(a_n))^{\perp\perp}.$$

This is not to be overstated: Gudder’s concrete logics [16] *are* set-representable non-Boolean orthoposets — but as point-ful posets, not via a  $\sigma$ -Loomis–Sikorski theorem.

## 5 The open problem

*Open Problem.* Does a *point-free*, non-distributive  $\sigma$ -additive probability theory exist? Concretely: is there a non-Boolean,  $\sigma$ -complete OML carrying a  $\sigma$ -additive probability that is genuinely *relational* — built from the entailment order of incompatible propositions, the non-Boolean analogue of a localic (point-free) measure — rather than reducible to a measure on an underlying point space? Or can one prove that any such structure collapses — either to a point space, or to the Boolean case?

“Point-free” qualifies the *measure*: it must not descend to a measure on a point space (the (B)-realisation of Remark 5.2), even when the lattice itself happens to be set-representable. The localic framing is forced, not preferred, because the point-based route is closed in advance. A full  $\sigma$ -tribe / Loomis–Sikorski representation preserves  $\wedge, \vee$  as  $\cap, \cup$ , so its image is a sublattice of a Boolean algebra, hence distributive — forcing the OML Boolean (Remark 2.8). So “relational probability via a faithful set representation” is impossible by a one-line argument for *any* non-Boolean OML; the only coherent target is the point-free one, where the measure lives on the lattice (or its dual frame) without descending to points. The two richly non-distributive poles frame the difficulty:  $L(H)$ +Gleason gives a  $\sigma$ -additive measure on the lattice but is *not* point-free — it is the *most* point-ful object available (Remark 5.2, the (A)/(B) analysis) — while the concrete combinatorial objects are *segregated* (non-distributivity confined to finite blocks, the

countable structure Boolean across them, §5.1). The open problem is whether a genuinely point-free, genuinely non-distributive  $\sigma$ -additive probability exists between these poles; a positive answer is the non-Boolean analogue of localic measure theory, a negative one an impossibility theorem closing the cluster.

### 5.1 What the combinatorial route settles, and why it is not enough

It is tempting to look for the object among concrete  $\sigma$ -orthocomplete OMLs, on the grounds that a state's  $\sigma$ -additivity invokes  $\bigvee_n a_n$  only for orthogonal families (Definition 2.5). This route is **settled, and it does not produce the object**. The property that would make  $\sigma$ -additivity genuinely exceed finite additivity on a concrete non-Boolean lattice is the interleaving condition

$$(\star) \quad \exists p, \exists \{a_n\} \text{ infinite orthogonal, } p \wedge a_n = 0 \ (\forall n), \quad p \not\leq a_n \ (\forall n), \quad (2)$$

and  $(\star)$  is satisfied *trivially* by the plain product  $\prod_n \mathbf{MO}_2$ : take  $a_n = a$  in coordinate  $n$  (zero elsewhere), and  $p = \bigvee_n b_n$  where  $b_n = b$  in coordinate  $n$  (zero elsewhere) — a genuine countable *orthogonal* join, so  $p = (b, b, b, \dots)$  exists by  $\sigma$ -orthocompleteness. Then  $p \wedge a_n = 0$  (coordinate  $n$ :  $a \wedge b = 0$  in  $\mathbf{MO}_2$ ; elsewhere  $a_n = 0$ ) and  $p \not\leq a_n$  (coordinate  $n$ :  $b \not\leq a^\perp$ ). Since  $\prod_n \mathbf{MO}_2$  is concrete,  $\sigma$ -complete (a product of finite, hence complete, lattices), non-Boolean, and infinite, it satisfies  $(\star)$  and *every* combinatorial hypothesis one might impose — yet it is exactly the *segregated* object the open problem must exclude. So the combinatorial axioms ( $\sigma$ -completeness, concreteness,  $(\star)$ ) are *not* the discriminator; the missing ingredient is non-segregation, which is a point-free notion, not a closure property.

*Remark 5.1* (The witness has no points, and the Boolean boundary). The universal closure above (a faithful set representation forces distributivity) already rules out the point-based route for every non-Boolean OML. The witness  $\prod_n \mathbf{MO}_2$  shows the failure concretely and at the level of points:  $\mathbf{MO}_2$  admits *no* two-valued *homomorphism* — each of its four separating two-valued *states* ( $a \mapsto 0, b \mapsto 0$ , etc.) fails  $\omega(a \vee b) \leq \omega(a) + \omega(b)$  since  $a \vee b = \mathbf{1}$ , so none preserves  $\vee$  — and  $\prod_n \mathbf{MO}_2$  inherits this, since its diagonal copy  $\{\mathbf{0}, \mathbf{1}, (a, a, \dots), (a^\perp, \dots), (b, b, \dots), (b^\perp, \dots)\}$  is a sub- $\mathbf{MO}_2$  on which any homomorphism would restrict to one of  $\mathbf{MO}_2$ 's (none exists). (This is special to  $\mathbf{MO}_2$  and its products — a generic non-Boolean OML such as  $\mathbf{2} \times \mathbf{MO}_2$  *does* carry homomorphisms, e.g. the projection; the universal obstruction is the distributivity argument, not homomorphism-scarcity.) So the witness is concrete (Definition 2.7: order-determining two-valued *states*) and  $\sigma$ -complete yet has no points in the homomorphism sense at all — the point-free target is forced, not engineered. The same arithmetic records the sharp Boolean boundary: an OML is Boolean iff it carries a unital set of *subadditive* states (Pták–Pulmannová [32]), so concrete non-Boolean OMLs exist precisely *because* their separating two-valued states fail subadditivity — and the forcing property is subadditivity, not  $\sigma$ -additivity.

A worked instance fixes the picture. Navara's construction [29] produces an infinite,  $\sigma$ -orthocomplete, rich OML whose inner block is a finite *stateless* Greechie logic — which is exactly what makes Navara's  $\mathcal{L}$  non-concrete. Replacing the stateless block by  $\mathbf{MO}_2$  (write  $\mathcal{L}_2$ ) removes the only source of non-concreteness: the assembly is a product of a sublogic of a Kalmbach horizontal sum (blocks glued only at  $\{0, 1\}$ , *not* an atom-sharing Greechie loop), and pure horizontal sums of concrete logics are concrete, as are sublogics and products. So  $\mathcal{L}_2$  is concrete,  $\sigma$ -orthocomplete, non-Boolean, and carries  $(\star)$  — yet it is not a new object: it is  $\prod_n \mathbf{MO}_2$  with extra scaffolding, segregated in the same way, and it likewise admits no  $\sigma$ -tribe representation. It is the cautionary example showing that exhibiting  $(\star)$  and concreteness together is easy and does *not* reach the open problem.

The art constrains the open ( $\sigma$ -complete) problem from both sides without closing it: *no* impossibility theorem covers the concrete class (Remark 4.1), while the two *theorem-closed* routes to an engine — RDP and MacNeille (§4.3(i),(ii)) — are ruled out, leaving only the third,  $\sigma$ -Stone duality, which is itself open. What is required is not a new idea in the abstract but a

representation in that gap: non-distributive enough to evade routes (i),(ii), yet  $\sigma$ -faithful where the known dualities are merely finitary.

*Remark 5.2* (Why  $L(H)$ +Gleason does *not* already settle this — the (A)/(B) point-space equivocation). “Probability from relations, point-free” may *look* delivered by  $L(H)$ +Gleason. It is not: the appearance rests on conflating two point-spaces, and *realisation* is defined as concentration on (B).

|     | Point-space            | Gleason’s relation to it                                                        |
|-----|------------------------|---------------------------------------------------------------------------------|
| (A) | Hilbert rays of $H$    | <i>requires</i> them — $\rho$ in $s = \text{tr}(\rho \cdot)$ is built from rays |
| (B) | MB dual filters $P(A)$ | <i>removes</i> them, but only by being a representation theorem                 |

Gleason is thus the *most* point-ful result available (every lattice state  $\mapsto$  the point datum  $\rho$ ). What  $L(H)$  exhibits is a *separation*, not point-freeness:

|                         |                                                                                          |
|-------------------------|------------------------------------------------------------------------------------------|
| on the lattice          | $\sigma$ -additive measures exist (Gleason, normal states);                              |
| on the dual $S_0(L(H))$ | no state extends to a charge (Prop. 3.2),<br>so a fortiori none concentrates on $P(A)$ . |

It cannot serve as the existence proof required, so the point-free  $\sigma$ -additive structure of §5 is the only candidate engine.

## 6 Directions

With the extension axis closed on  $L(H)$  (Proposition 3.2) and the combinatorial ( $\sigma$ -orthocomplete, set-representable) route shown to deliver only segregated objects (§5.1), the live problem is the point-free existence question of §5. Three directions bear on it.

### Build the point-free engine, or prove it collapses

Seek a  $\sigma$ -additive probability on a non-Boolean  $\sigma$ -complete OML (or its dual frame) that does *not* descend to a point space — a genuine non-Boolean localic measure — or prove that any such structure collapses to a point space or to the Boolean case. The point-based form is already ruled out (Remark 5.1: non-Boolean OMLs have no two-valued homomorphisms, so no  $\sigma$ -tribe representation exists at any cardinality); the open content is precisely whether the point-free form can be non-trivial, which is why the engine must be sought locally (route (iii), §4.3), not as a field of sets.

### Settle the (A)/(B) status of $L(H)$ +Gleason

Decide whether  $L(H)$ +Gleason counts as a genuine point-free non-distributive  $\sigma$ -additive witness or is disqualified by the point-space equivocation (Remark 5.2); if it counts, ask whether it is essentially unique. This is the same problem as the first item under a semantic rather than constructive description — “does point-free non-distributive  $\sigma$ -additive probability exist?” is the content of the localic-engine question — so progress on either informs the other.

### Sharpen the impossibility side via subadditivity

The Boolean boundary is subadditivity of the separating states (Pták–Pulmannová, Remark 5.1). A negative resolution would likely run through

the dual frame: quantify how far a non-Boolean OML’s measures must fail subadditivity, and whether requiring a point-free  $\sigma$ -additive measure forces enough subadditive structure to collapse the object to a point space or to the Boolean case — the impossibility form of the open problem.

A candidate direction now closed: the singular case for  $L(H)$ . One might have hoped a singular state would extend where no normal state does, providing the first explicit “extends but may fail to concentrate” example; Proposition 3.2 rules this out, catching every state, singular included. The reduction is recorded because the singular case was expected to require infinitary methods and does not (Remark 6.1).

*Remark 6.1* (Finite versus countable: finite suffices). Is the obstruction to extending a singular state realised by a *finite* or only by a *countable* pairwise-meet-zero family? The two arguments divide on exactly this point:

**line argument (Rmk 3.3)**      finitary, but **vacuous** on singular ( $s(e) = 0 \ \forall$  finite  $e$ )  
**MO<sub>2</sub>,  $\infty$ -dim atoms (Prop. 3.2)**    finite ( $n = 4$ ),  $\sum_i s(a_i) = 2$  state-independently

So the singular obstruction is *not* entangled with the descent-axis passage, as the vacuity of the line argument might suggest: it is realised by a finite family, hence a self-contained finitary problem, separable from the general construction and not governed by the finitary-to- $\sigma$  passage.

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